```
    {
        return $this->carriages * 30;
    }
}

$train = new Train(6);
print "This train can carry {$train->getMaxPassengers()} people" . PHP_EOL;
```

```
Output:
This train can carry 180 people
```

Here we have a `Train` class, which now implements a `Vehicle` interface. `Vehicle` specifies that a derived class must implement `getMaxPassangers()`, and the method must return an integer. Imagine that our previously derived vehicles implement a fixed value (cars still carry five people and so on), but trains are a special case: their carrying capacity is variable according to the number of carriages they have (determined when the object is constructed in this case).

With inheritance-based OCP, we'd have a problem: using a parent class of `Vehicle` wouldn't meet the needs of `Train` objects. We either have to add logic to the parent implementation or overload the parent, and not every derived class needs a `carriages` property. By only closing the interface, we are free to work with different implementations for different derived classes. This does, of course, raise the possible issue of violating the Don't Repeat Yourself (DRY) principle, but common implementations can be composed from a `trait` or the use of a Strategy pattern (where the required functionality is injected via another object), which we will look at later on in the chapter.

## Liskov Substitution Principle (LSP)

Barbara Liskov presented a paper – *Data Abstraction and Hierarchy* – at a keynote address of the OOPSLA (Object-Oriented Programming Systems, Languages and Applications) conference in 1987. Her work, together with Jeannette Wing, concerned the data relationships between parent and child types and focused on their substitutability. The LSP concerns *strong behavioral subtyping*: namely that any child class should satisfy the expectations of clients that are written to use a parent class. Furthermore, the child class substitution should not just be safe (in that it causes no syntax errors, i.e., all parent members are found in the child), but it should behave correctly too.